Branch/Class/Batch: AI&DS/T.E/A1
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EXPERIMENT NUMBER: 1

AIM: .
To study the allocation of the electromagnetic spectrum and understand the wireless frequency bands used in communication systems.

OBJECTIVES: .

1. To understand the electromagnetic spectrum and its characteristics.
2. To study the allocation of different frequency bands.
3. To identify wireless communication bands and their applications.
4. To understand the importance of spectrum allocation in avoiding interference.
5. To learn the frequency ranges used in modern wireless communication technologies.

THEORY:

Introduction

The **electromagnetic (EM) spectrum** is the complete range of electromagnetic waves arranged according to their frequency or wavelength. These waves are used for various communication systems such as radio broadcasting, television, mobile communication, satellite communication, Wi-Fi, Bluetooth, and radar.

The electromagnetic spectrum is divided into different frequency bands. Governments regulate these bands through organizations such as the **International Telecommunication Union (ITU)** and national regulatory bodies (e.g., WPC in India) to prevent interference between communication systems.

Wireless Frequency Bands

Band	Frequency Range	Applications
VLF (Very Low Frequency)	3–30 kHz	Submarine communication
LF (Low Frequency)	30–300 kHz	Navigation systems
MF (Medium Frequency)	300 kHz–3 MHz	AM Radio
HF (High Frequency)	3–30 MHz	Shortwave radio
VHF (Very High Frequency)	30–300 MHz	FM Radio, TV Broadcasting
UHF (Ultra High	300 MHz–3	Television, Mobile Communication,

Frequency)	GHz	GPS
SHF (Super High Frequency)	3–30 GHz	Wi-Fi, Radar, Satellite Communication, 5G
EHF (Extremely High Frequency)	30–300 GHz	Millimeter Wave Communication, Advanced 5G

Wireless Bands Used in Communication

1. Bluetooth

- Frequency Band: **2.4 GHz ISM Band**
- Range: 10–100 m
- Used for wireless headphones, keyboards, smart devices.

2. Wi-Fi

- **2.4 GHz**
- **5 GHz**
- **6 GHz (Wi-Fi 6E/7)**
- Used for wireless internet connectivity.

3. Cellular Networks

Generation	Frequency Bands
2G	900 MHz, 1800 MHz
3G	850 MHz, 900 MHz, 2100 MHz
4G LTE	700 MHz, 800 MHz, 1800 MHz, 2300 MHz
5G	600 MHz–6 GHz and 24–100 GHz (mmWave)

4. GPS

- Frequency: **1.227 GHz and 1.575 GHz**
- Used for navigation and positioning.

5. Satellite Communication

- L Band (1–2 GHz)
- S Band (2–4 GHz)
- C Band (4–8 GHz)
- X Band (8–12 GHz)
- Ku Band (12–18 GHz)
- Ka Band (26–40 GHz)

6. RFID

- Low Frequency (125–134 kHz)
- High Frequency (13.56 MHz)
- UHF (860–960 MHz)

Applications include access cards, inventory management, and toll collection.

Advantages of Spectrum Allocation

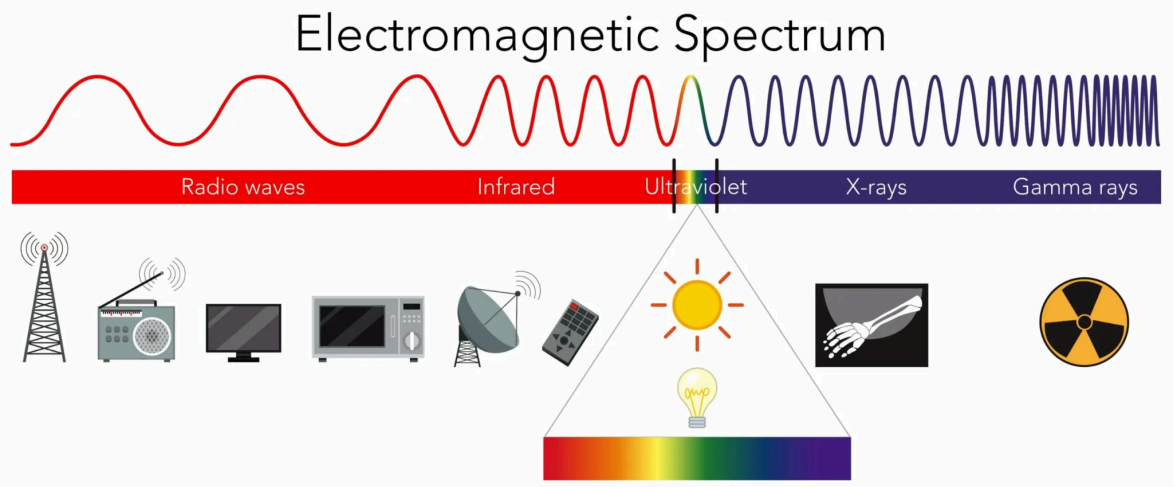
- Prevents interference between communication systems.
- Enables efficient use of available frequencies.
- Supports reliable wireless communication.
- Ensures compatibility among different communication technologies.
- Improves communication quality and network performance.

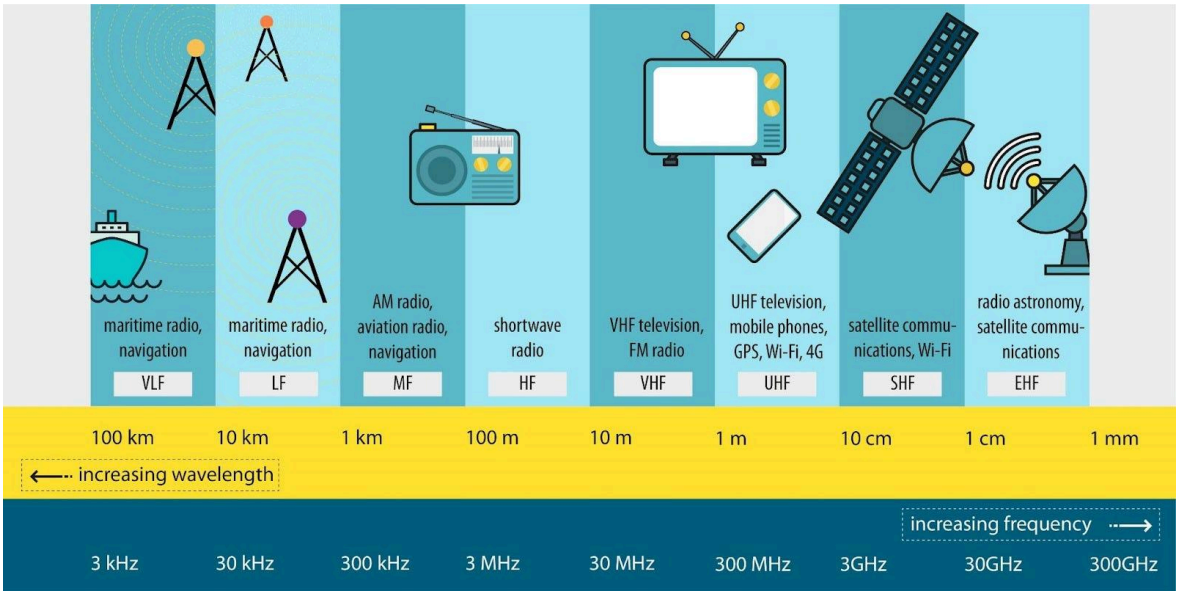
Applications

- Mobile communication
- Television broadcasting
- Radio broadcasting
- Wi-Fi and Bluetooth
- Satellite communication
- GPS navigation
- Radar systems
- Military communication
- Internet of Things (IoT)

CONCULSION:
The electromagnetic spectrum is an essential resource for wireless communication. Proper spectrum allocation ensures efficient utilization of frequency bands, minimizes interference, and enables reliable operation of modern communication technologies such as Wi-Fi, Bluetooth, cellular networks, satellite communication, GPS, and radar. Understanding these frequency bands is fundamental for designing and managing wireless communication systems.

OUTPUT:




maritime radio, navigation
VLF
maritime radio, navigation
LF
AM radio, aviation radio, navigation
MF
shortwave radio
HF
VHF television, FM radio
VHF
UHF television, mobile phones, GPS, Wi-Fi, 4G
UHF
satellite communications, Wi-Fi
SHF
radio astronomy, satellite communications
EHF
100 km
10 km
1 km
100 m
10 m
1 m
10 cm
1 cm
1 mm
increasing wavelength
increasing frequency
3 kHz
30 kHz
300 kHz
3 MHz
30 MHz
300 MHz
3GHz
30GHz
300GHz